DrivR-Base: A Feature Extraction Toolkit for Variant Effect Prediction Model Construction

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Abstract

Motivation: Recent advancements in sequencing technologies have led to the discovery of numerous variants in the human genome. However, understanding their precise roles in diseases remains challenging due to their complex functional mechanisms. Various methodologies have emerged to predict the pathogenic significance of these genetic variants. Typically, these methods employ an integrative approach, leveraging diverse data sources that provide critical insights into genomic function. Despite the abundance of publicly available data sources and databases, the process of navigating, extracting, and pre-processing features for machine learning models can be daunting. Furthermore, researchers often invest substantial effort in feature extraction, only to later discover that these features lack informativeness.

Results: In this paper, we present *DrivR-Base*, an innovative resource that efficiently extracts and integrates molecular information (features) for single nucleotide variants from a wide range of databases and tools, including AlphaFold, ENCODE, and *Variant Effect Predictor (VEP)*. The resulting features can be used as input for machine learning models designed to predict the pathogenic impact of human genome variants in disease. Moreover, these feature sets have applications beyond this, including haploinsufficiency prediction and the development of drug repurposing tools. We describe the resource's development, practical applications, and potential for future expansion and enhancement.

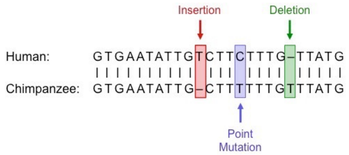
Availability and Implementation: *DrivR-Base* source code is available *at https://github.com/amyfrancis97/DrivR-Base*.

Introduction

- Rapid evolution of next-generation sequencing technologies has unveiled extensive variants within the human genome.
- Many variants are still classified as variants of uncertain significance, prompting the need for understanding their role in diseases.
- Distinguishing driver variants from neutral ones will be crucial for understanding disease mechanisms.
- Diverse machine learning methodologies, including *AlphaMissense*[1], *FATHMM-MKL*[2], and *CADD*[3], have been developed for this purpose.
- Though model architecture and underlying training data remain distinct, these tools share a common phenomenon in that they heavily rely on integrated feature data for accurate predictions.
- Here, we present *DrivR-Base*, a novel repository designed to streamline data acquisition for robust variant driver predictors.
- The repository extracts information for ten distinct feature groups, incorporating under-explored data such as DNA shape properties and structural information from AlphaFold.

1. DNA Sequence Conservation

Ref. 4

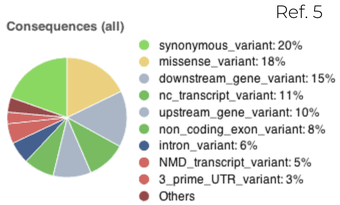


Metrics include PhyloP and PhastCons scores, Umap, and Bismap mappability data, assessing nucleotide substitution rates, and sequence mapping accuracy.

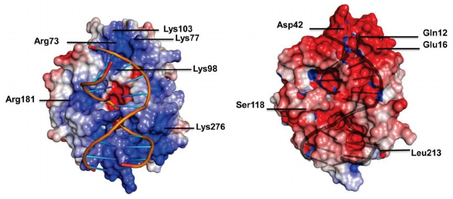
2. Variant Effect Predictor

Ref. 5

Organized into three groups, *DrivR-Base* extracts predicted transcript consequences, wild-type and mutant amino acids, and distances to transcripts for each variant.



3. Dinucleotide Properties

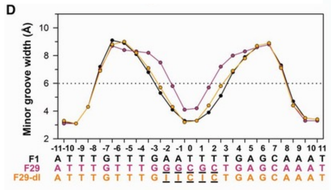


Ref. 6

Sourced from DiProDB, this dataset comprises 125 conformational and thermodynamic dinucleotide properties for both wild-type and mutant configurations.

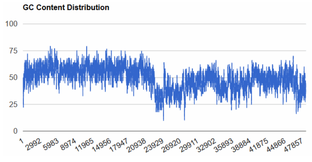
4. DNA Shape

Ref. 7



Incorporating five DNA shape properties from DNAShapeR, it calculates features like minor groove width, helix twist, and electrostatic potential for genomic regions around a variant.

5. GC Content & CpG Sites



Ref. 8

DrivR-Base calculates GC content, CpG counts, and observed versus expected CpG ratios for nine window sizes.

6. Kernel-Based Sequence Similarity

Utilizing p-spectrum kernels, we capture disruptions in sequences flanking a single nucleotide variant, assessing the composition of k-mers in wild-type and mutant sequences..

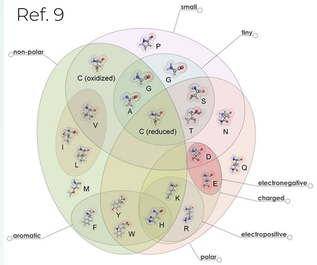
$$\Phi_u^p(s) = |\{(v_1, v_2) : s = v_1 \mathrm{u}\, v_2\}| \qquad K_p(s,t) = \sum_{u \in \sum p} \Phi_u^p(s)\Phi_u^p(t)$$

7. Amino Acid Sequence Conservation

Extracting substitution rates from various matrices for non-synonymous variants, including PAM and BLOSUM matrices.

8. Amino Acid Properties

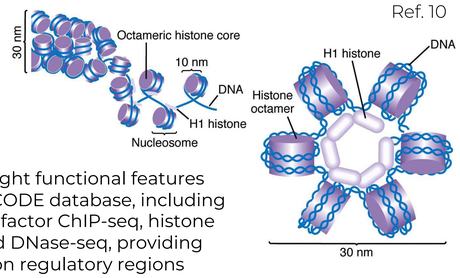
Ref. 9



532 amino acid properties are extracted for both wild-type and mutant amino acid sequences from AAindex in R. These features encompass characteristics such as polarity, hydrophobicity, local flexibility, and helix-bend preferences.

9. ENCODE Regulatory Features

Ref. 10



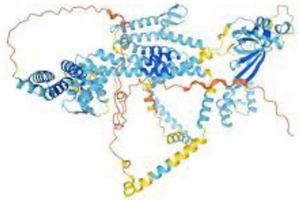
We extract eight functional features from the ENCODE database, including transcription factor ChIP-seq, histone ChIP-seq, and DNase-seq, providing information on regulatory regions flanking a variant.

10. Alpha Fold Structural Features

DrivR-Base integrates structural data from AlphaFold and PDB for coding variants, extracting X, Y, and Z atom coordinates, isotropic atomic displacement parameters (IADP), and structural conformation types.

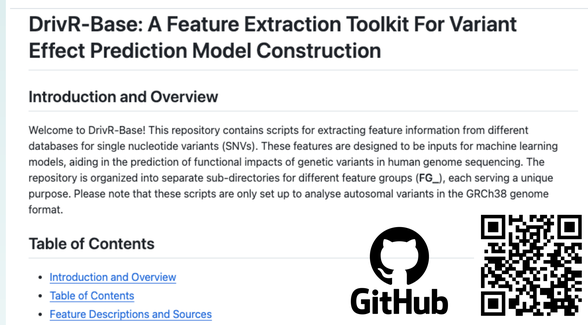
Ref. 11

Ref. 12



DrivR-Base Implementation

The *DrivR-Base* repository provides a comprehensive toolkit of scripts, documentation, and links to original sources, facilitating the construction of a cohesive set of features for variant interpretation.



DrivR-Base: A Feature Extraction Toolkit For Variant Effect Prediction Model Construction

Introduction and Overview

Welcome to DrivR-Base! This repository contains scripts for extracting feature information from different databases for single nucleotide variants (SNVs). These features are designed to be inputs for machine learning models, aiding in the prediction of functional impacts of genetic variants in human genome sequencing. The repository is organized into separate sub-directories for different feature groups (FG_), each serving a unique purpose. Please note that these scripts are only set up to analyse autosomal variants in the GRCh38 genome format.

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Conclusions

DrivR-Base is a versatile cross-database extraction toolkit designed for the analysis of human single nucleotide variants, streamlining the data acquisition process for researchers in this field.

Diverse Applications of Feature Sets:
- Construction of high-dimensional machine-learning models for variant driver status prediction.
- Prediction of haploinsufficiency.
- Identification of functional similarities to known drug targets, potentially facilitating drug repurposing efforts.

Future Outlooks:
- Expansion of capabilities to encompass a broader range of mutations, including indels, deletions, and structural rearrangements.
- Diversification of available feature groups for extraction.

Collaboration and Feedback:
- Researchers are encouraged to contact the authors for discussions on the incorporation of additional feature groups or enhancing existing ones within *DrivR-Base*.

Acknowledgements & References

This work was funded by Cancer Research UK [C18281/A30905] & carried out in the UK Medical Research Council Integrative Epidemiology Unit (MC_UU_00032/03), University of Bristol.

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